WORKED SOLUTION

STEP II 2009, Question 13

Probability and Statistics • Binomial Expected Cost

STEP II 2009, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s2-q13

The key idea

The number of engines that fail is binomial, so the repair cost is a simple function of a binomial random variable, and its expectation is a short sum. The trick for reaching the neat form $(*)$ is to avoid summing the "replace everything" cases one by one: use the complement, and then factor $1-q^n = p(1+q+\cdots+q^{n-1})$ to pull out the factor of p.

Solution

We assume, as the question intends, that engines fail independently of one another, each with probability p, and write $q = 1-p$. All costs are measured in units of K until the end.

The Andover

Let X be the number of Andover engines that fail, so $X \sim \mathrm{B}(4,p)$, and let C_A be the repair cost. Then

$$C_A = \begin{cases} 0 & X = 0, \\ K & X = 1, \\ 4K & X \geq 2. \end{cases}$$

Now $\mathrm{P}(X=0) = q^4$ and $\mathrm{P}(X=1) = 4pq^3$, so $\mathrm{P}(X \geq 2) = 1 - q^4 - 4pq^3$. Hence

$$\begin{aligned} \mathrm{E}(C_A) &= K \cdot 4pq^3 + 4K(1-q^4-4pq^3) \\ &= 4K(1-q^4) - 12Kpq^3. \end{aligned}$$

Since $1-q^4 = (1-q)(1+q+q^2+q^3) = p(1+q+q^2+q^3)$,

$$\begin{aligned} \mathrm{E}(C_A) &= 4Kp(1+q+q^2+q^3) - 4Kp \cdot 3q^3 \\ &= 4Kp(1+q+q^2-2q^3), \end{aligned}$$

which is $(*)$, as required.

The Basingstoke

Let $Y \sim \mathrm{B}(6, p)$ be the number of Basingstoke engines that fail, and C_B the repair cost. Then $C_B = YK$ when $Y \le 2$ and $C_B = 6K$ when $Y \ge 3$. With

$$\mathrm{P}(Y=0) = q^6, \qquad \mathrm{P}(Y=1) = 6pq^5, \qquad \mathrm{P}(Y=2) = 15p^2q^4,$$

we get

$$\begin{aligned}\mathrm{E}(C_B) &= K\cdot 6pq^5 + 2K\cdot 15p^2q^4 + 6K(1-q^6-6pq^5-15p^2q^4)\\ &= 6K(1-q^6) - 30Kpq^5 - 60Kp^2q^4.\end{aligned}$$

Using $1-q^6 = p(1+q+q^2+q^3+q^4+q^5)$ and taking out $6Kp$,

$$\mathrm{E}(C_B) = 6Kp(1+q+q^2+q^3+q^4+q^5-5q^5-10pq^4).$$

> **Ask yourself** The bracket still contains a p. How do we make it "a form similar to $(*)$"?

Form $(*)$ is a factor of p times a polynomial in q alone, so we replace the remaining p by $1-q$: $10pq^4 = 10q^4 - 10q^5$. Collecting terms,

$$\boxed{\mathrm{E}(C_B) = 6Kp(1+q+q^2+q^3-9q^4+6q^5).}$$

When is the Andover cost $\frac{2}{3}$ of the Basingstoke cost?

We need $\mathrm{E}(C_A) = \frac{2}{3}\mathrm{E}(C_B)$, that is

$$4Kp(1+q+q^2-2q^3) = 4Kp(1+q+q^2+q^3-9q^4+6q^5).$$

Both sides have the factor $4Kp$, and $K>0$, so this holds exactly when $p=0$ or

$$-2q^3 = q^3-9q^4+6q^5 \iff 3q^3-9q^4+6q^5=0 \iff 3q^3(1-q)(1-2q)=0.$$

So $q=0$, $q=1$ or $q=\frac{1}{2}$, giving $p=1$, $p=0$ or $p=\frac{1}{2}$. All three lie in $[0,1]$, so the required values are

$$p=0, \qquad p=\tfrac{1}{2}, \qquad p=1.$$

The value $p=0$ is the dull one: no engine ever fails, and both expected costs are zero. At $p=1$ every engine fails, costing $4K$ and $6K$. The interesting value is $p=\frac{1}{2}$.

Checks

At $p=\frac{1}{2}$, $(*)$ gives $\mathrm{E}(C_A) = 2K(1+\frac{1}{2}+\frac{1}{4}-\frac{1}{4}) = 3K$, and the Basingstoke formula gives $\mathrm{E}(C_B) = 3K(1+\frac{1}{2}+\frac{1}{4}+\frac{1}{8}-\frac{9}{16}+\frac{6}{32}) = \frac{9}{2}K$, and indeed $3 = \frac{2}{3}\cdot\frac{9}{2}$. We also checked both formulae exactly against direct enumeration over the binomial distribution for 200 rational values of p, and by simulation: at $p=0.3$, 2×10^5 simulated launches gave average costs $1.811K$ and $2.483K$, against the formulae's $1.805K$ and $2.485K$.

Common slips

- Charging XK rather than $4K$ when two or more Andover engines fail: when the threshold is passed, *all* engines are replaced, including the good ones.
- Leaving a p inside the Basingstoke bracket, so the answer isn't in the form asked for.
- Dividing by p in the final equation and losing $p = 0$, or dividing by q^3 and losing $p = 1$.

Beyond STEP

This is a small example of expected cost under a threshold policy, the bread and butter of reliability theory and operational research: a manager chooses the threshold to balance the cost of replacing good parts against the cost of more frequent repairs. The tail-sum manipulation we used, writing $\mathrm{P}(X \geq k)$ as one minus the first few terms, is the same idea that gives $\mathrm{E}(X) = \sum_{k \geq 1} \mathrm{P}(X \geq k)$ for non-negative integer variables. Grimmett and Welsh, *Probability: An Introduction*, develops these ideas carefully.

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